

Dr. Beici Liang

Last updated: July, 2026

Email: liangbeici@gmail.com

Website: beiciliang.github.io

16975 Stockholm

Sweden

- Machine Learning Engineer with 10+ years building audio and ML systems end-to-end
- Ships production ML at scale: Python, SQL, Docker, Kubernetes, DevOps 
- PhD in music tech, and a popular-science writer making MIR accessible to the public 

Working Experience

Aug. 2024 – **Senior Machine Learning Engineer**

now

[Epidemic Sound](#), Sweden

Lead DRM algorithm design to automate music activation and rights claiming across UGC platforms. Spearheaded the shift of our music recognition technology to a vector-based approach powered by representation learning, driving a significant increase in DRM revenue.

Sep. 2023 – **Senior Backend and Cloud Engineer**

July. 2024

[Nomono AS](#), Norway

Responsible for the backend development of [Nomono Cloud](#) to build a robust, scalable and high-performing cloud service that provides great audio processing and editing for content creators.

May 2021 – **Head of R&D**

Aug. 2023

Music Tech Startups including [SPARWK AS](#) and [Deus Vault UK Ltd.](#)

Developed AI algorithms for music entity linking system and audio-based information retrieval services, e.g., genre detection, tempo estimation, etc.; Filed 4 patents for the systems and methods used in artists and repertoire (A&R).

Sept. 2019 – **Senior Research Engineer**

April 2021

[Tencent Music Entertainment \(TME\)](#), China

Developed end-to-end AI models for music auto-tagging, structural segmentation, large scaled singer recognition, and audio embeddings for music recommendation; Provided a better understanding of the music content for over 20 million tracks, and benefited over 800 million users in China via the QQ Music App; Published 5 conference papers and 3 patents, and awarded with the Annual Technology Breakthrough.

Education

2014 – 2019 **PhD in Media and Arts Technology**

School of Electronic Engineering and Computer Science

Queen Mary University of London (QMUL), United Kingdom

Research Group: Centre for Digital Music ([C4DM](#))

Supervisors: Mark Sandler, George Fazekas, Andrew McPherson

Thesis: [Modelling Instrumental Gestures and Techniques - A Case Study of Piano Pedalling](#)

2010 – 2014 **BEng in Integrated Circuit Design and Integrated System**

School of Electronic Information Engineering

Tianjin University ([TJU](#)), China

Grade: 88/100

Open-source Projects

2018 – now	intro2musictech Introduce music technology to Chinese audiences and build MIR communities in China. 12k+ followers on Zhihu and 2k+ subscribers on WeChat Official Account.	
2023	aws-bootcamp-cruddur-2023 Implementations for a micro-blogging platform using React, Flask and AWS.	
2019	sustain-pedal-detection Python implementations for piano sustain pedal detection.	
2018	modelAttackDecay-for-piano-transcription Python implementations of an attack/decay model for piano transcription.	

Miscellaneous

Volunteers

- Scientific Program Chair of the 24th International Society for Music and Information Retrieval Conference ([ISMIR 2023](#))
- Mentor and volunteer for Women in Music Information Retrieval ([WiMIR](#))
- Member of the Local Organising Committee for the 12th International Audio Mostly Conference
- Reviewers and memberships in numerous international conferences and journals such as ISMIR, IEEE Signal Processing Society, Audio Engineering Society, etc.

Main Publications

More can be seen at [Google Scholar](#).

2021	K. Chen, Liang, B. , X. Ma, and M. Gu. “Learning Audio Embeddings with User Listening Data for Content-Based Music Recommendation”. In: <i>2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)</i> . pp. 3015-3019. doi: 10.1109/ICASSP39728.2021.9414458 .	
2021	S. Hu, Liang, B. , Z. Chen, X. Lu, E. Zhao, and S. Lui. “Large-Scale Singer Recognition Using Deep Metric Learning: An Experimental Study”. In: <i>2021 International Joint Conference on Neural Networks (IJCNN)</i> . pp. 1–6. doi: 10.1109/IJCNN52387.2021.9533911 .	
2020	S. Hu, B. Zhang, Liang, B. , E. Zhao, and S. Lui. “Phase-Aware Music Super-Resolution Using Generative Adversarial Networks”. In: <i>Interspeech 2020</i> . pp. 4074–4078. doi: 10.21437/Interspeech.2020-2605 .	
2019	Liang, B. , G. Fazekas, and M. Sandler. “Transfer Learning for Piano Sustain-Pedal Detection”. In: <i>2019 International Joint Conference on Neural Networks (IJCNN)</i> . pp. 1-6. doi: 10.1109/ijcnn.2019.8851724 .	 
2019	Liang, B. , G. Fazekas, and M. Sandler. “Piano Sustain-Pedal Detection Using Convolutional Neural Networks”. In: <i>2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)</i> . pp. 241-245. doi: 10.1109/ICASSP.2019.8683505 .	 
2018	Liang, B. , G. Fazekas, and M. Sandler. “Measurement, Recognition, and Visualization of Piano Pedalling Gestures and Techniques”. <i>Journal of the Audio Engineering Society</i> 66.6 (2018), pp. 448-456. doi: 10.17743/jaes.2018.0035 .	